

Designing Technology for an Aging Population

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ABSTRACT

The population of the developed world is aging. Most websites, apps, and digital devices are used by adults aged 50+ as well as by younger adults, so they should be designed accordingly. This one-part course, based on the presenter's recent book, presents age-related factors that affect older adults' ability to use digital technology, as well as design guidelines that reflect older adults' varied capabilities, usage patterns, and preferences.

CCS CONCEPTS

• **Social and professional topics** → User characteristics; Age; Seniors; User characteristics; People with disabilities; • **Human-centered computing** → Accessibility; Accessibility design and evaluation methods; Interaction design; Interaction design process and methods; User interface design; Interaction design; Interaction design process and methods; Participatory design.

KEYWORDS

older adults, user interface, user experience, design guidelines, design heuristics

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1 BENEFITS

The world's population is aging, especially in developed countries. The audience for most websites, apps, and digital devices includes older adults, so most websites, apps, and digital devices should be designed to accommodate them. Based on our experience designing, reviewing, and testing apps and websites with older adults, and on an extensive review of the literature, this course describes age-related factors that affect older adults' ability to use digital technology, and present design guidelines that reflect older adults' varied capabilities, usage patterns, and preferences.

This course is appropriate for CHI because it presents material that more HCI practitioners should know in order to be able to produce universally usable desktop and mobile apps and websites. It is especially appropriate for CHI 2021 because Japan – the conference's pre-COVID-19 host country – is the country with the fastest-growing proportion of older adults.

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1.1 Learning Objectives

In this course, participants will learn:

- that the developed world's population is rapidly aging, including the statistics supporting that fact,
- about age-related factors that affect the usability of digital user interfaces,
- how poorly-designed apps, devices, and websites can present barriers for older people,
- guidelines for designing in an inclusive way, so technology can accommodate older adults,
- that this problem will not disappear as today's older adults die off; it will persist indefinitely, because technology continually advances, while people tend to get "stuck" in the technology-era in which they grow up,
- special considerations and methods for designing and evaluating digital products and services with older adults.

2 INTENDED AUDIENCE

This course is intended for software designers and developers of all levels of experience. Others who might benefit: Software Q/A engineers, usability testers, and development managers.

3 PREREQUISITES

None, however some experience designing user interfaces would be helpful.

4 CONTENT

This class covers:

- demographics of users of digital technology, by age
- why the need to design for older adults will not go away when today's older adults are gone,
- age-related factors affecting ability to use computers, mobile devices, and the Web,
- common design problems that decrease usability for older adults,
- design guidelines to help designers improve experiences for technology users of all ages.

4.1 Agenda

- Introduction and definitions
- Demographics of digital technology usage, by age
- The problem: too many apps and websites that are unfriendly to older adults
- Motivation for solving the problem
- Age-related changes and characteristics
- Guidelines for age-friendly technology
- Small-group exercise: review website or app for usability by older adults
- Summary, wrap-up, evaluations

5 PRACTICAL WORK

Small-group exercises reviewing the usability of websites and apps by older adults. This can be done using Zoom breakout rooms if the course is presented online.

6 INSTRUCTOR BACKGROUND

Jeff Johnson is a Professor in the Computer Science Department of the University of San Francisco. After earning B.A. and Ph.D. degrees from Yale and Stanford, he worked at Cromemco, Xerox, US West, Hewlett-Packard Labs, and Sun Microsystems. In 1990, he co-chaired the first Participatory Design conference, PDC'90. He serves on the SIGCHI U.S. Public Policy Committee. He has also taught at Stanford University and Mills College, and in 2006

and 2013 he taught HCI as an Erskine Fellow at the University of Canterbury in New Zealand. He is a member of the ACM SIGCHI Academy, a SIGCHI Lifetime Achievement in Practice Awardee, and an ACM Distinguished Member. He has authored or co-authored many articles and chapters on Human-Computer Interaction, as well as the books *GUI Bloopers*, *Web Bloopers*, *GUI Bloopers 2.0*, *Designing with the Mind in Mind*, *Conceptual Models* (co-authored with Austin Henderson), *Designing with the Mind in Mind*, 2nd edition, *Designing UIs for an Aging Population* (co-authored with Kate Finn), and *Designing with the Mind in Mind*, 3rd edition.

REFERENCES

- [1] Johnson, J. and Finn, K. (2017) *Designing User Interfaces for an Aging Population: Towards Universal Design*, Morgan Kaufmann Publisher.